

CROP Physiology

Diverging Responses in Transpiration of Soybean Genotypes to Vapor Pressure Deficit

Internship Program - Visiting Scholar
From March to December 2024

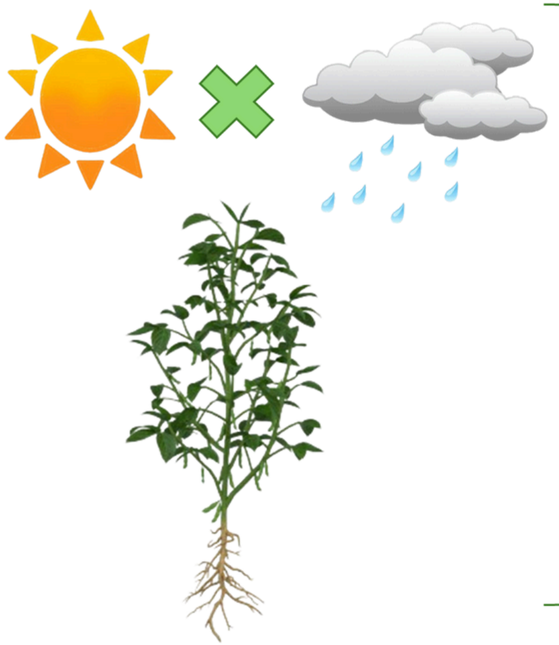
Advisor: Elvis Elli
Student: Paula Andrade Pereira
e-mail: paulaandrad72@gmail.com

UNITED SOYBEAN BOARD

UNIVERSITY OF ARKANSAS

UofA DIVISION OF AGRICULTURE RESEARCH & EXTENSION
University of Arkansas System

INTRODUCTION & OBJECTIVES



Water use eficiency (WUE)

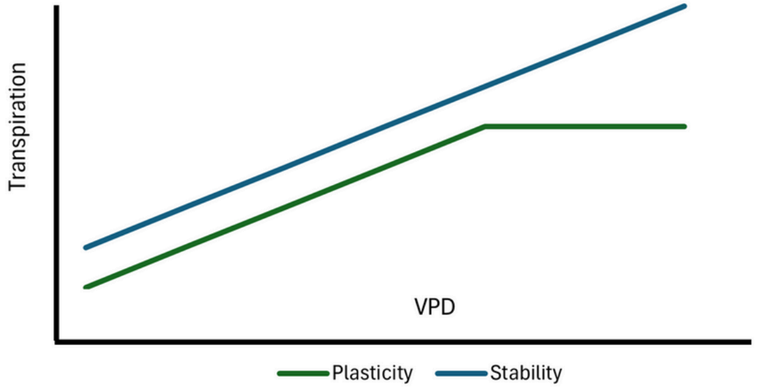
↓

Alter transpiration rates in response to vapor pressure Deficit (VPD)

↓

Plasticity vs Stability

Hypothesis



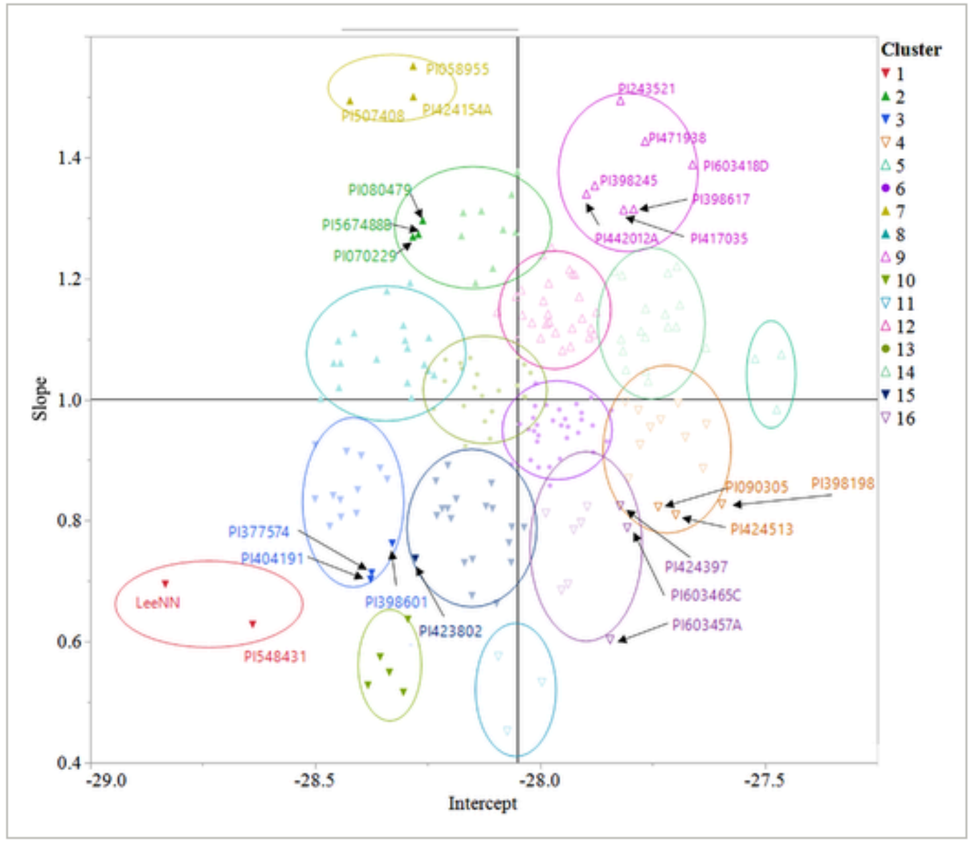
METHODOLOGY

Experiment Arrangment

- Select the genotypes;

  Q1: Plastic + High WUE;
  Q2: Plastic + Low WUE;
  Q3: Nonplastic + Low WUE;
  Q4: Nonplastic + High WUE;

- Seed treatment.



Abstract reference:

Chamarthi, S.K., Kaler, A.S., Abdel-Haleem, H., Fritschi, F.B., Gillman, J.D., Ray, J.D., Smith, J.R., Purcell, L.C., 2022. Identification of genomic regions associated with the plasticity of carbon 13 ratio in soybean. Plant Genome e20284.

METHODOLOGY

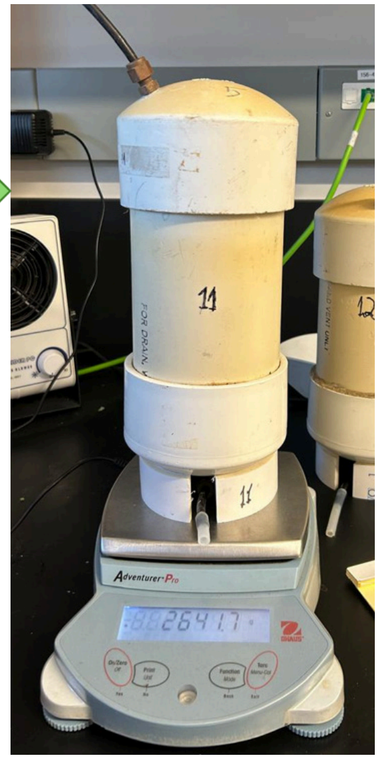
Organizing the pots



1 pot net



Rocks + 2 pot nets



Tare for each pot

METHODOLOGY

Preparing the pots



Substrate



Fill the pots and add water 3 times



Drain the pots until there is no water coming out of the hose

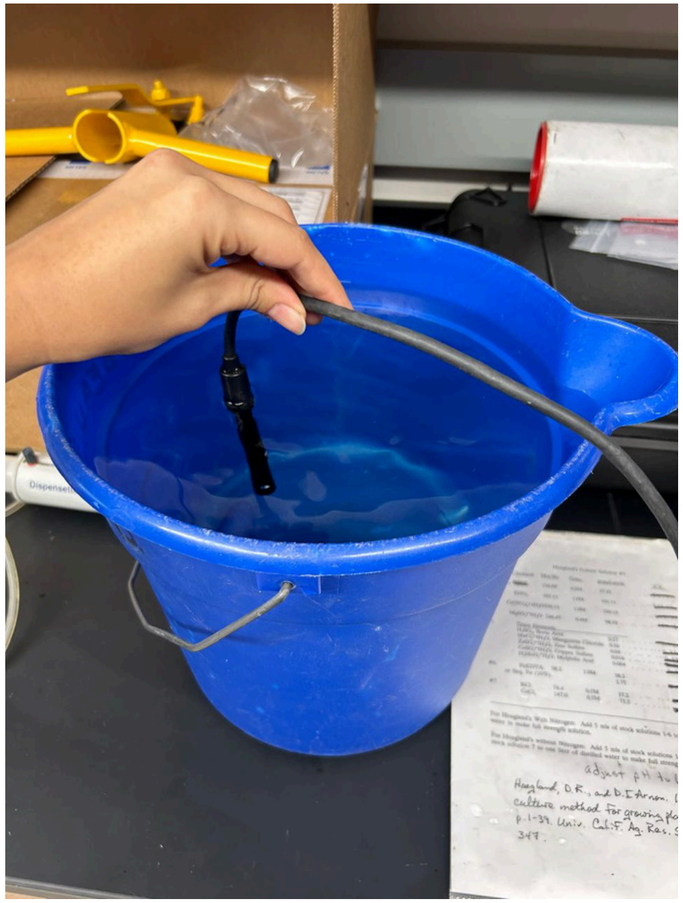


Drain all the pots

For part 2 and 3: 90% substrate and 10% soil

METHODOLOGY

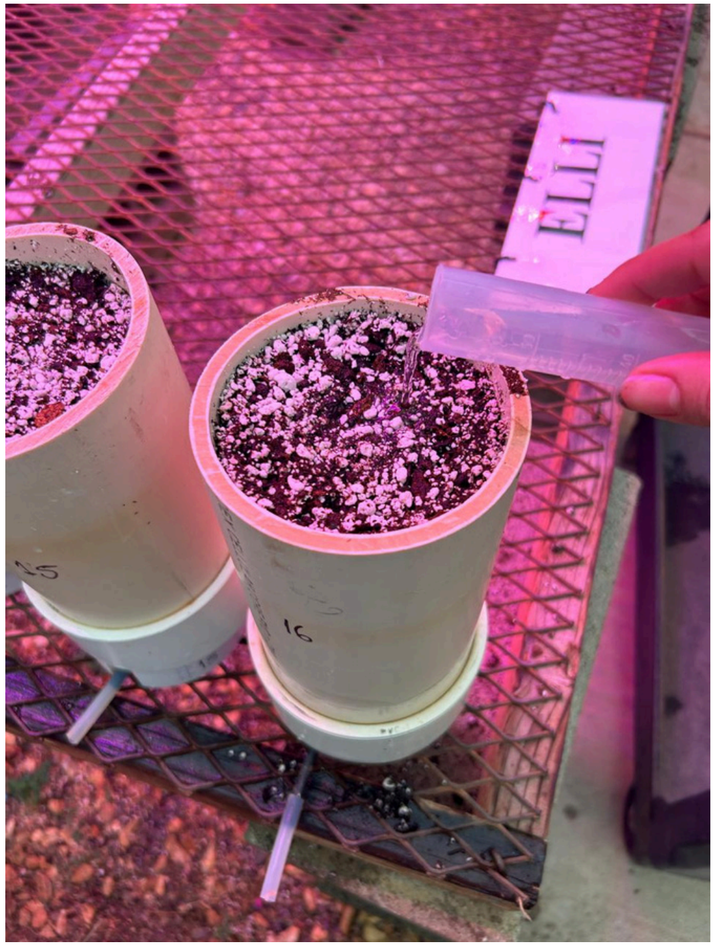
Hoagland's Culture Solution 1 - without Nitrogen



- Quantities indicated to prepare 1 liter;
- PH target: 6.8;
  - NaOH ➡ Increases
  - HCl ➡ Decreases
- Total added/substrate:
  - 100% substrate - 500ml/pot
  - 90% substrate - 450ml/pot



METHODOLOGY



Adding Hoagland's Culture Solution to pots

- Split the total amount, adding 50ml each time, avoiding saturation. Reserve 50ml to be added after the seeds are planted;
- Ex: 500 ml in total -> Add 450 ml before planting: 50 ml x 9 times;
- Once it is done, fill the pots with substrate until the top.

METHODOLOGY

Planting

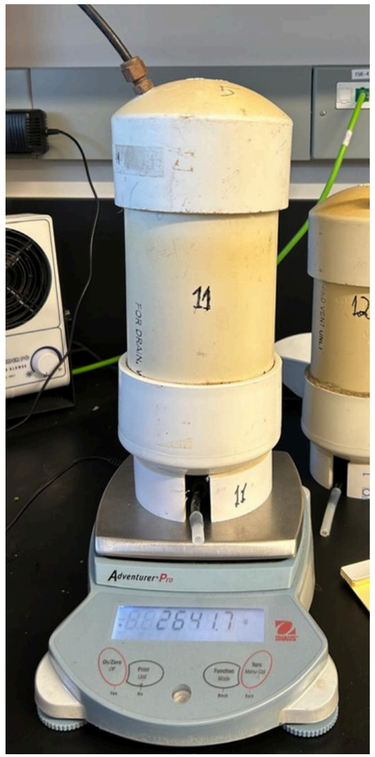
- 6 seeds/pot -> thinning;
- Add Rhizobium;
- Add the final 50ml of the solution;
- Let the pots overnight in the green house and weight them early next morning.





METHODOLOGY

Watering method until R1



"Final weight"

-

Tare

=

100%

pot capacity

Watering target:

50 - 70% pot capacity



VPD TREATMENTS

Part 1

05/15 and 05/16



Part 2

10/30 and 10/31



Part 3

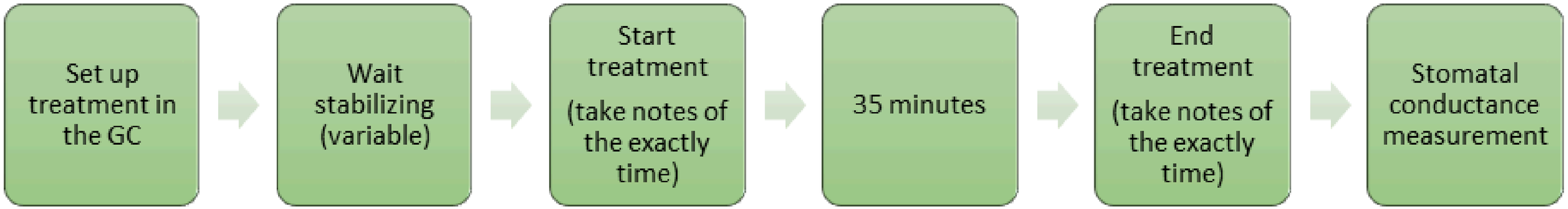
11/07 and 11/08



VPD TREATMENTS - PART 1

Part 1

05/15 - Day 1



All the pots were watered up to 70% of the pot capacity

Afternoon of the previous day

All the pots were watered between T3 and T4

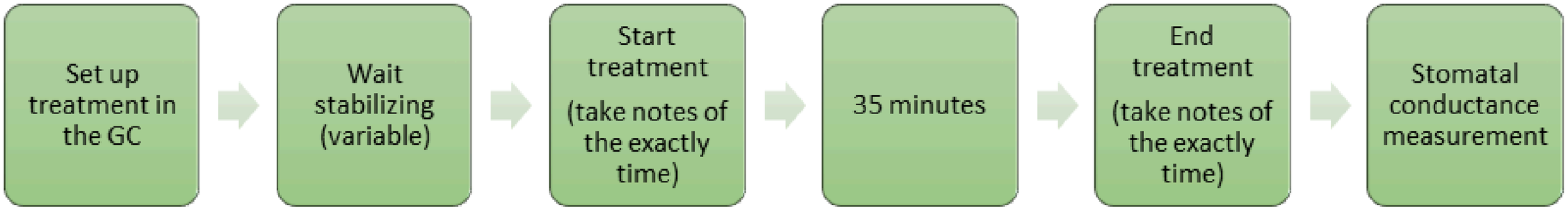
6 VPD treatments

The scales were used

VPD TREATMENTS - PART 1

Part 1

05/16 - Day 2

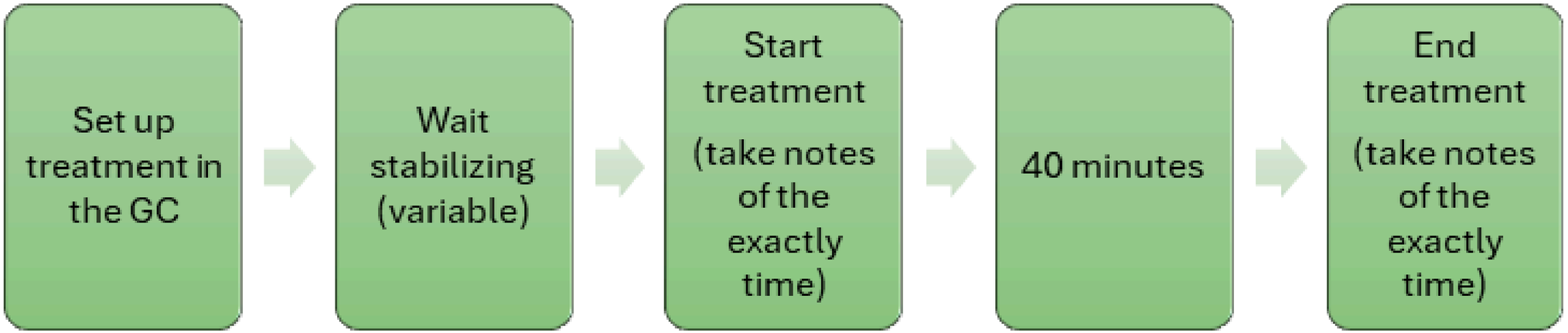


The pots were watered up to 70%
1 hour before of the beginning of the treatments → 6 VPD treatments → The scales were used

VPD TREATMENTS - PART 2

Part 2

10/30 - 10/31

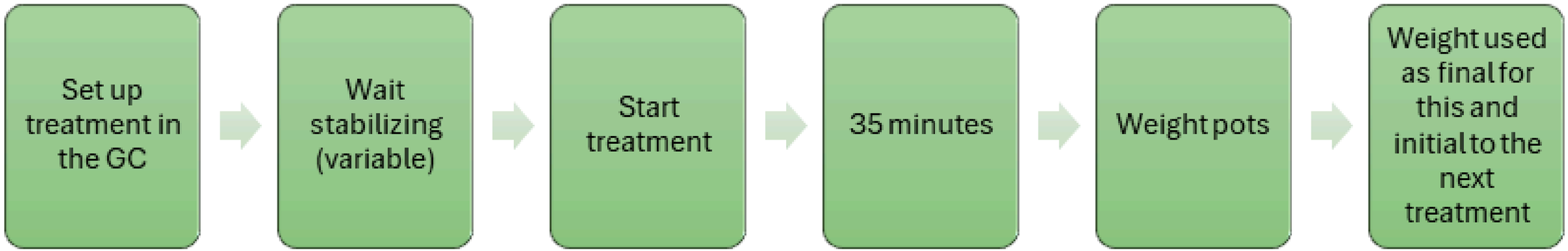


Set up the GC to T1 the afternoon before → The pots were watered up to 70% 1 hour before of the beginning of the treatments → 6 VPD treatments → The scales were used

VPD TREATMENTS - PART 3

Part 3

11/07 - Day 1



Set up the GC to T1 the afternoon before → The pots were watered up to 70% 1 hour before of the beginning of the treatments → 5 VPD treatments → The scales were not used

VPD TREATMENTS - PART 3

Part 3

11/07 - Day 1

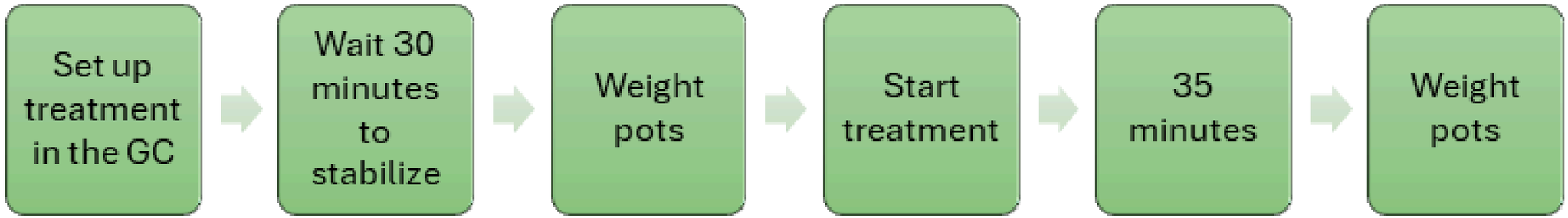
Weight used as final for this and initial to the next treatment → Variable time gaps between treatments → Adjust treatments durations

T2:
Time used to GC stabilize between T1 and T2 + 35 minutes

VPD TREATMENTS - PART 3

Part 3

11/08 - Day 2



Set up the GC to T1 the afternoon before → The pots were watered up to 70% 1 hour before of the beginning of the treatments → 5 VPD treatments → The scales were not used

FOLIAR AREA



Partitioning

↓

Foliar area

↓

Dry mass

↓

Data analysis

DATA ANALYSIS

Transpiration (ml) → With scales

↓

Relative Transpiration

↓

Transpiration per leaf area

↓

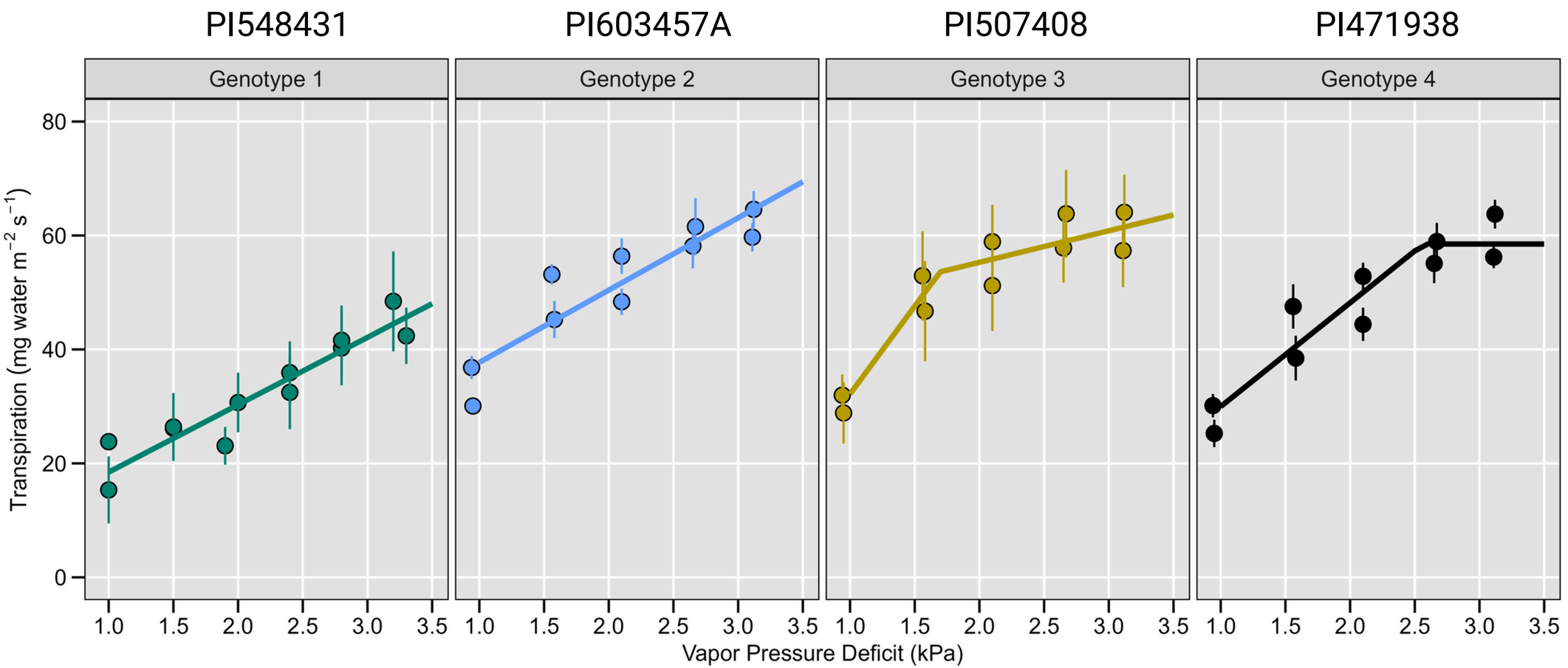
Pots + genotype average

↓

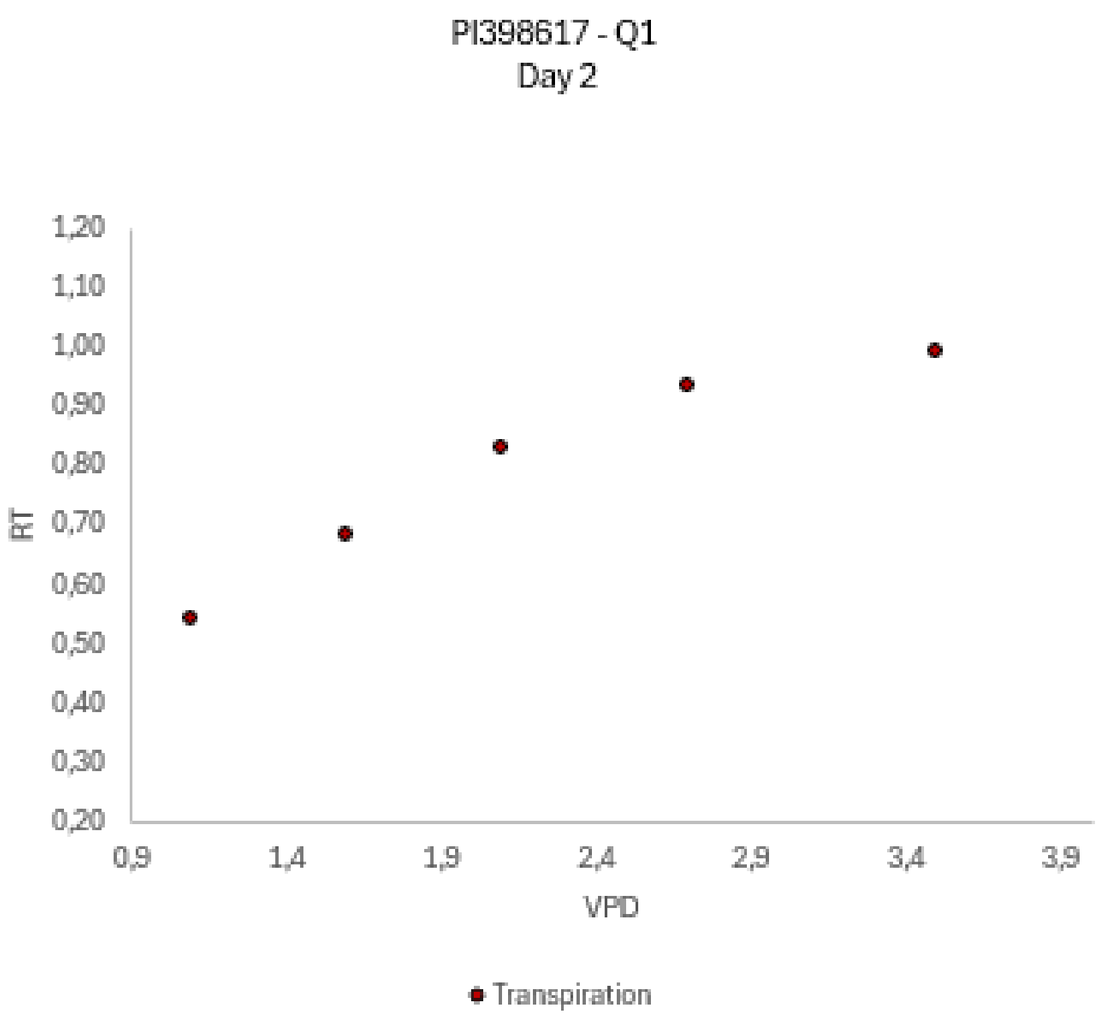
Average of the 3 first and last measurements

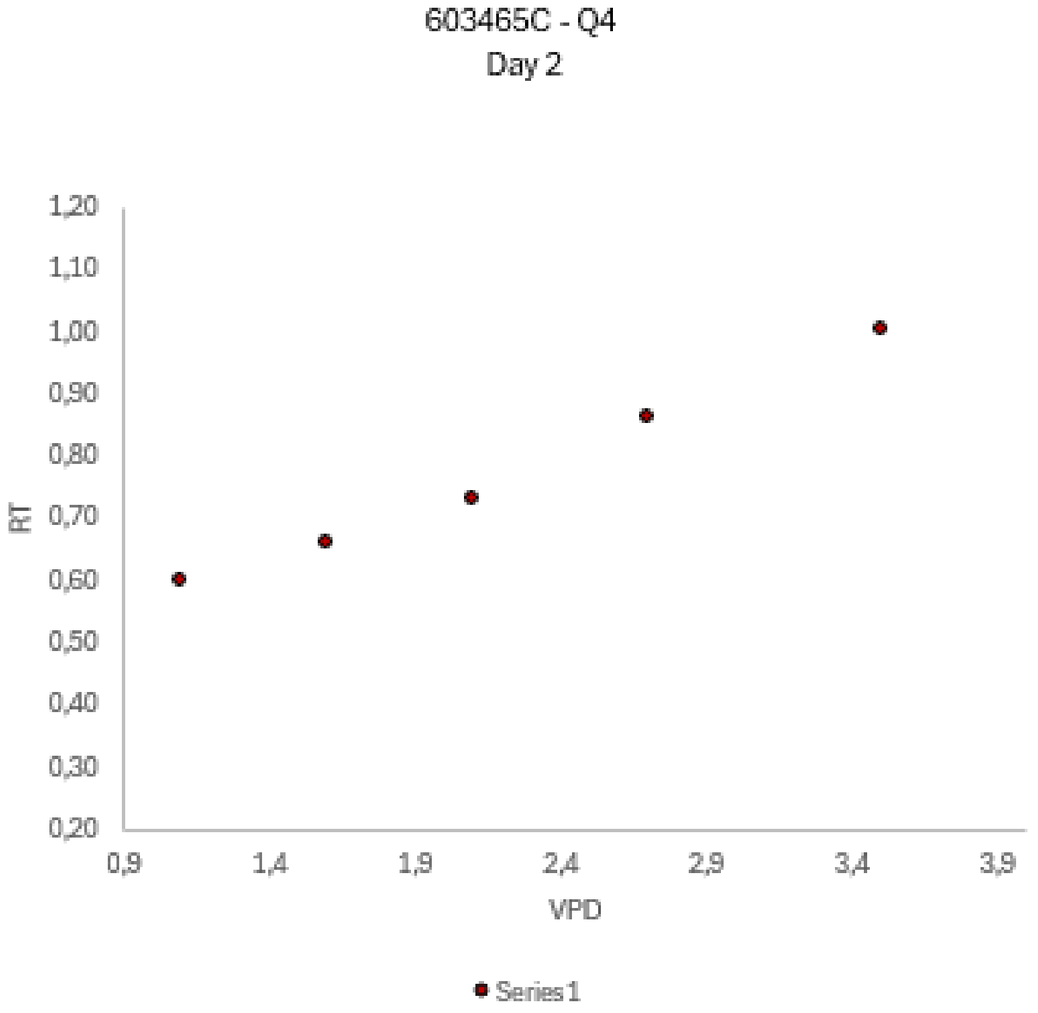
15:16	10/31		642	4	3340	9500		233557	250731	232377	244717	265248	258294	225778	232800	232358	233411	227780				
15:18	10/31		642	6	3360	9500		233451	250632	232254	244601	265153	258179	225685	232708	232288	233292	227680				
15:20	10/31		642	8	3370	9500		233351	250518	232155	244508	265066	258086	225623	232592	232204	233206	227585				
15:22	10/31		642	10	3370	9500		233242	250419	232036	244393	264985	258014	225532	232499	232118	233120	227510				
15:24	10/31		642	12	3390	9500		233121	250306	231937	244287	264902	257900	225449	232400	232041	233028	227446				
15:26	10/31		642	14	3390	9500		233026	250210	231839	244172	264834	257812	225377	232295	231962	232912	227309				
15:28	10/31		642	16	3380	9500		232892	250098	231745	244059	264740	257722	225308	232194	231882	232820	227228				
15:30	10/31		642	18	3360	9500		232795	249997	231640	243963	264666	257645	225225	232107	231768	232694	227158				
15:32	10/31		642	20	3360	9500		232675	249904	231559	243855	264571	257592	225160	232030	231664	232603	227049				
15:34	10/31		642	22	3360	9500		232550	249795	231455	243757	264480	257516	225069	231913	231570	232501	226955				
15:36	10/31		642	24	3360	9500		232427	249675	231396	243647	264404	257455	224988	231837	231465	232404	226846				
15:38	10/31		642	26	3360	9500		232320	249591	231280	243548	264335	257363	224908	231742	231379	232319	226751				
15:40	10/31		642	28	3360	9500		232202	249498	231169	243454	264250	257329	224829	231660	231304	232233	226675				
15:42	10/31		642	30	3370	9500		232074	249388	231069	243337	264170	257234	224760	231567	231230	232180	226612				
15:44	10/31		642	32	3390	9500		231973	249312	230974	243235	264104	257176	224674	231472	231146	232052	226498				
15:46	10/31		642	34	3390	9500		231865	249210	230890	243135	264035	257091	224606	231378	231081	231935	226438				
15:48	10/31		642	36	3400	9500		231750	249107	230795	243033	263966	257022	224539	231319	231018	231846	226379				
15:50	10/31		642	38	3390	9500		231636	249030	230696	242933	263897	256908	224463	231239	230948	231756	226288				
15:52	10/31		642	40	3390	9500		231522	248920	230617	242844	263801	256861	224391	231125	230854	231667	226244				
15:54	10/31		642	42	3350	9500		231393	248822	230545	242765	263731	256835	224303	231071	230749	231581	226140				
15:56	10/31		642	44	3330	9500		231291	248722	230469	242653	263636	256784	224207	231013	230620	231522	226008				
15:58	10/31		642	46	3310	9500		231174	248626	230406	242577	263547	256752	224122	230919	230510	231386	225904				

RESULTS


PI548431
PI603457A
PI507408
PI471938
Genotype 1
Genotype 2
Genotype 3
Genotype 4
Transpiration (mg water m^{-2} s^{-1})
0
20
40
60
80
1.0
1.5
2.0
2.5
3.0
3.5
Vapor Pressure Deficit (kPa)

RESULTS

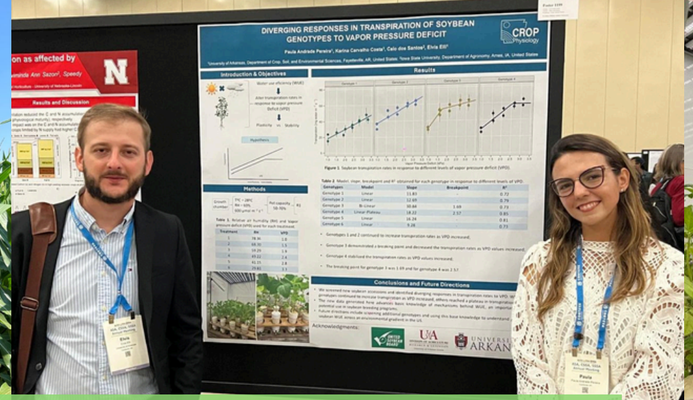

PI398617 - Q1
Day 2
1,20
1,10
1,00
0,90
0,80
0,70
0,60
0,50
0,40
0,30
0,20
RT
0,9
1,4
1,9
2,4
2,9
3,4
3,9
VPD
Transpiration


603465C - Q4
Day 2
1,20
1,10
1,00
0,90
0,80
0,70
0,60
0,50
0,40
0,30
0,20
RT
0,9
1,4
1,9
2,4
2,9
3,4
3,9
VPD
Series1

WHAT I LEARNED

- As less as we open the GC door during the treatments is better;
- It may be necessary to “help the seeds to germinate”;
- The GC takes, at least 15 minutes to start being close to the target, avoid keep looking at it before that;
- Use a regular time to let it stabilize may be better (ex: 30 minutes);
- If it is not possible to use the automatic scales, weight the pots inside of the GC;
- Take notes of the exactly time for everything (normal and GC time).





THANK YOU, CROP PHYSIOLOGY LAB! ♡









